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LABSHEET-4

1.Create a class Student with instance fields name, age and a class variable count to track number of students. Use local variables to assign values and display details.

```
class Student {
  // Instance fields
  String name;
  int age;

  // Class variable
  static int count = 0;

  // Constructor
  Student(String name, int age) {
    // Local variables are used as parameters
    this.name = name;
    this.age = age;
    count++;
  }

  // Method to display student details
  void displayDetails() {
    System.out.println("Name: " + name);
    System.out.println("Age: " + age);
    System.out.println();
  }

  public static void main(String[] args) {
    // Creating student objects
```

```
    Student s1 = new Student("Aryan Tomar", 18);
    Student s2 = new Student("Riya", 19);

    // Display details
    s1.displayDetails();
    s2.displayDetails();

    // Display total number of students
    System.out.println("Total Students: " + count);
  }
}
```

OUTPUT:-

Name: Aryan Tomar

Age: 18

Name: Riya

Age: 19

Total Students: 2

2) Create a class Rectangle with instance fields length, breadth, a class variable shapeName = "Rectangle". Write methods with local variables to calculate and display the area.

```
class Rectangle {
  // Instance fields
  double length;
  double breadth;

  // Class variable
  static String shapeName = "Rectangle";

  // Method to calculate and display area
```

```
    void displayArea() {
        // Local variable
        double area = length * breadth;

        System.out.println("Shape: " + shapeName);
        System.out.println("Length: " + length);
        System.out.println("Breadth: " + breadth);
        System.out.println("Area: " + area);
    }

    public static void main(String[] args) {
        // Create object
        Rectangle r = new Rectangle();

        // Assign values
        r.length = 10;
        r.breadth = 5;

        // Display area
        r.displayArea();
    }
}
```

OUTPUT:-

```
Shape: Rectangle
Length: 10.0
Breadth: 5.0
Area: 50.0
```

3) Create a class Circle with instance field radius, a class variable PI = 3.14. Use local variables to calculate area and circumference.

```
class Circle {
  // Instance field
  double radius;

  // Class variable
  static double PI = 3.14;

  // Method to calculate area and circumference
  void calculate() {
    // Local variables
    double area = PI * radius * radius;
    double circumference = 2 * PI * radius;

    System.out.println("Radius: " + radius);
    System.out.println("Area: " + area);
    System.out.println("Circumference: " + circumference);
  }

  public static void main(String[] args) {
    // Create object
    Circle c = new Circle();

    // Assign radius
    c.radius = 5;

    // Calculate and display
```

```
    c.calculate();
  }
}
```

OUTPUT:-

Radius: 5.0

Area: 78.5

Circumference: 31.400000000000002

4) Create a class Employee with instance fields emp Id, salary, and a class variable company Name. Display employee details using local variables.

```
class Employee {
  // Instance fields
  int empId;
  double salary;

  // Class variable
  static String companyName = "ABC Company";

  // Method to display employee details
  void displayDetails() {
    // Local variables
    int id = empId;
    double sal = salary;

    System.out.println("Employee ID: " + id);
    System.out.println("Salary: " + sal);
    System.out.println("Company Name: " + companyName);
  }
```

```
    public static void main(String[] args) {
        // Create object
        Employee e = new Employee();

        // Assign values
        e.empId = 101;
        e.salary = 35000;

        // Display details
        e.displayDetails();
    }
}
```

OUTPUT:-

Employee ID: 101

Salary: 35000.0

Company Name: ABC Company

5) Create a class Book with fields title, author, price, and a class variable publisher. Use local variables to set and display book details.

```
class Book {
    String title;
    String author;
    double price;

    static String publisher = "ABC Publications";

    void displayDetails() {
        String t = "Java Programming";
        String a = "James Gosling";
```

```
    double p = 450;

    title = t;
    author = a;
    price = p;

    System.out.println("Title: " + title);
    System.out.println("Author: " + author);
    System.out.println("Price: " + price);
    System.out.println("Publisher: " + publisher);
  }

  public static void main(String[] args) {
    Book b = new Book();
    b.displayDetails();
  }
}
```

OUTPUT:-

Title: Java Programming

Author: James Gosling

Price: 450.0

Publisher: ABC Publications

6) Create a class Car with fields brand, mileage, and a class variable wheels = 4. Display details

```
class Car {
  String brand;
  double mileage;

  static int wheels = 4;
```

```
  void displayDetails() {
    String b = "Toyota";
    double m = 20.5;

    brand = b;
    mileage = m;

    System.out.println("Brand: " + brand);
    System.out.println("Mileage: " + mileage + " km/l");
    System.out.println("Wheels: " + wheels);
  }

  public static void main(String[] args) {
    Car c = new Car();
    c.displayDetails();
  }
}
```

OUTPUT:-

Brand: Toyota

Mileage: 20.5 km/l

Wheels: 4

7) Create a class Bank Account with fields account Number, balance, and a class variable bank Name. Use local variables in deposit method.

```
class BankAccount {
  String accountNumber;
  double balance;
```

```
    static String bankName = "State Bank";

    void deposit() {
        double amount = 5000;

        balance = balance + amount;

        System.out.println("Bank Name: " + bankName);
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Deposit Amount: " + amount);
        System.out.println("Balance: " + balance);
    }

    public static void main(String[] args) {
        BankAccount b = new BankAccount();

        b.accountNumber = "1234567890";
        b.balance = 10000;

        b.deposit();
    }
}
```

OUTPUT:-

```
Bank Name: State Bank
Account Number: 1234567890
Deposit Amount: 5000.0
Balance: 15000.0
```

8) Create a class Pen with fields color, type, and a class variable manufacturer. Use local variables to display pen details.

```
class Pen {
   String color;
   String type;

   static String manufacturer = "Cello";

   void displayDetails() {
      String c = "Blue";
      String t = "Ball Pen";

      color = c;
      type = t;

      System.out.println("Color: " + color);
      System.out.println("Type: " + type);
      System.out.println("Manufacturer: " + manufacturer);
   }

   public static void main(String[] args) {
      Pen p = new Pen();
      p.displayDetails();
   }
}
```

OUTPUT:-

```
Color: Blue
Type: Ball Pen
```

Manufacturer: Cello

9) Create a class Laptop with fields brand, RAM, and a class variable os = "Windows". Use local variables to display details.

```
class Laptop {
   String brand;
   int RAM;

   static String os = "Windows";

   void displayDetails() {
      String b = "Dell";
      int r = 8;

      brand = b;
      RAM = r;

      System.out.println("Brand: " + brand);
      System.out.println("RAM: " + RAM + " GB");
      System.out.println("Operating System: " + os);
   }

   public static void main(String[] args) {
      Laptop l = new Laptop();
      l.displayDetails();
   }
}
```

OUTPUT:-

Brand: Dell

RAM: 8 GB

Operating System: Windows

10) Create a class Movie with fields name, genre, rating, and a class variable industry. Display movie details using local variables.

```
class Movie {
  String name;
  String genre;
  double rating;

  static String industry = "Bollywood";

  void displayDetails() {
    String n = "3 Idiots";
    String g = "Comedy-Drama";
    double r = 8.4;

    name = n;
    genre = g;
    rating = r;

    System.out.println("Movie Name: " + name);
    System.out.println("Genre: " + genre);
    System.out.println("Rating: " + rating);
    System.out.println("Industry: " + industry);
  }

  public static void main(String[] args) {
    Movie m = new Movie();
```

```
        m.displayDetails();
    }
}
```

11) Create a class Calculator with methods for add, subtract, multiply, divide using local variables. Add a class variable operations Count to count total operations done.

```
class Calculator {
    static int operationsCount = 0;

    void add() {
        int a = 20;
        int b = 10;
        int result = a + b;
        operationsCount++;
        System.out.println("Addition: " + result);
    }

    void subtract() {
        int a = 20;
        int b = 10;
        int result = a - b;
        operationsCount++;
        System.out.println("Subtraction: " + result);
    }

    void multiply() {
        int a = 20;
        int b = 10;
        int result = a * b;
```

```
        operationsCount++;
        System.out.println("Multiplication: " + result);
    }

    void divide() {
        int a = 20;
        int b = 10;
        int result = a / b;
        operationsCount++;
        System.out.println("Division: " + result);
    }

    public static void main(String[] args) {
        Calculator c = new Calculator();

        c.add();
        c.subtract();
        c.multiply();
        c.divide();

        System.out.println("Total Operations: " + operationsCount);
    }
}
```

OUTPUT:-

Addition: 30

Subtraction: 10

Multiplication: 200

Division: 2

Total Operations: 4

12) Create a class Student Marks with fields marks1, marks2, marks3, and a class variable university Name. Use local variables to calculate average.

```
class StudentMarks {
  int marks1;
  int marks2;
  int marks3;

  static String universityName = "COER University";

  void calculateAverage() {
    int m1 = marks1;
    int m2 = marks2;
    int m3 = marks3;

    int total = m1 + m2 + m3;
    double average = total / 3.0;

    System.out.println("Marks 1: " + m1);
    System.out.println("Marks 2: " + m2);
    System.out.println("Marks 3: " + m3);
    System.out.println("University: " + universityName);
    System.out.println("Average: " + average);
  }

  public static void main(String[] args) {
    StudentMarks s = new StudentMarks();
```

```
        s.marks1 = 80;
        s.marks2 = 75;
        s.marks3 = 85;

        s.calculateAverage();
    }
}
```

OUTPUT:-

Marks 1: 80

Marks 2: 75

Marks 3: 85

University: COER University

Average: 80.0

13) Create a class Temperature with a field celsius and a class variable scale = "Celsius". Use local variables to convert into Fahrenheit and Kelvin.

```
class Temperature {
    double celsius;

    static String scale = "Celsius";

    void convertTemperature() {
        double c = celsius;
        double fahrenheit = (c * 9 / 5) + 32;
        double kelvin = c + 273.15;

        System.out.println("Temperature: " + c + " " + scale);
        System.out.println("Fahrenheit: " + fahrenheit);
```

```
    System.out.println("Kelvin: " + kelvin);
  }

  public static void main(String[] args) {
    Temperature t = new Temperature();

    t.celsius = 25;

    t.convertTemperature();
  }
}
```

OUTPUT:-

Temperature: 25.0 Celsius

Fahrenheit: 77.0

Kelvin: 298.15

14) Create a class Person with fields name, age and a class variable country. Use local variables to check voting eligibility.

```
class Person {
  String name;
  int age;

  static String country = "India";

  void checkEligibility() {
    String n = name;
    int a = age;

    System.out.println("Name: " + n);
```

```
    System.out.println("Age: " + a);
    System.out.println("Country: " + country);

    if (a >= 18) {
      System.out.println("Eligible for Voting");
    } else {
      System.out.println("Not Eligible for Voting");
    }
  }

  public static void main(String[] args) {
    Person p = new Person();

    p.name = "Aryan Tomar";
    p.age = 18;

    p.checkEligibility();
  }
}
```

OUTPUT:-

Name: Aryan Tomar

Age: 18

Country: India

Eligible for Voting

15) Create a class Simple Interest with fields principal, rate, time, and a class variable bank. Use local variables to calculate interest.

```
class SimpleInterest {
  double principal;
```

```
double rate;
double time;

static String bank = "State Bank";

void calculateInterest() {
   double p = principal;
   double r = rate;
   double t = time;

   double interest = (p * r * t) / 100;

   System.out.println("Bank: " + bank);
   System.out.println("Principal: " + p);
   System.out.println("Rate: " + r + "%");
   System.out.println("Time: " + t + " years");
   System.out.println("Simple Interest: " + interest);
}

public static void main(String[] args) {
   SimpleInterest s = new SimpleInterest();

   s.principal = 10000;
   s.rate = 5;
   s.time = 2;

   s.calculateInterest();
```

```
    }
}
```

OUTPUT:-

Bank: State Bank

Principal: 10000.0

Rate: 5.0%

Time: 2.0 years

Simple Interest: 1000.0

16) Create a class Triangle with sides a, b, c, and a class variable type "Scalene/Isosceles/Equilateral". Use local variables to check validity.

```
class Triangle {
    double a;
    double b;
    double c;

    static String type = "Scalene/Isosceles/Equilateral";

    void checkValidity() {
        double x = a;
        double y = b;
        double z = c;

        if (x + y > z && x + z > y && y + z > x) {
            System.out.println("Triangle is Valid");

            if (x == y && y == z) {
                System.out.println("Type: Equilateral");
            } else if (x == y || y == z || x == z) {
```

```
            System.out.println("Type: Isosceles");
        } else {
            System.out.println("Type: Scalene");
        }
    } else {
        System.out.println("Triangle is Not Valid");
    }
  }

  public static void main(String[] args) {
    Triangle t = new Triangle();

    t.a = 5;
    t.b = 6;
    t.c = 7;

    t.checkValidity();
  }
}
```

OUTPUT:-

Triangle is Valid

Type: Scalene

17) Create a class Student Grade with fields name, marks, and a class variable passing Marks. Use local variables to assign grade.

```
class StudentGrade {
  String name;
  int marks;
```

```
static int passingMarks = 40;

void assignGrade() {
   String n = "Aryan Tomar";
   int m = 85;

   name = n;
   marks = m;

   System.out.println("Name: " + name);
   System.out.println("Marks: " + marks);

   if (m >= passingMarks) {
      if (m >= 90) {
         System.out.println("Grade: A+");
      } else if (m >= 80) {
         System.out.println("Grade: A");
      } else if (m >= 70) {
         System.out.println("Grade: B");
      } else if (m >= 60) {
         System.out.println("Grade: C");
      } else {
         System.out.println("Grade: D");
      }
   } else {
      System.out.println("Grade: F");
   }
```

```
  }

  public static void main(String[] args) {
    StudentGrade s = new StudentGrade();
    s.assignGrade();
  }
}
```

OUTPUT:-

Name: Aryan Tomar

Marks: 85

Grade: A

18) Create a class Electric Bill with a field units and a class variable fixed Charge. Use local variables to calculate total bill.

```
class ElectricBill {
  int units;

  static double fixedCharge = 100;

  void calculateBill() {
    int u = units;
    double charge;

    if (u <= 100) {
      charge = u * 5;
    } else if (u <= 200) {
      charge = (100 * 5) + ((u - 100) * 7);
    } else {
      charge = (100 * 5) + (100 * 7) + ((u - 200) * 10);
```

```
        }

        double totalBill = charge + fixedCharge;

        System.out.println("Units: " + u);
        System.out.println("Fixed Charge: " + fixedCharge);
        System.out.println("Total Bill: " + totalBill);
    }

    public static void main(String[] args) {
        ElectricBill e = new ElectricBill();

        e.units = 250;

        e.calculateBill();
    }
}
```

OUTPUT:-

Units: 250

Fixed Charge: 100.0

Total Bill: 1800.0

19) Create a class Product with fields id, name, price, and a class variable discount Rate. Use local variables to calculate final price.

```
class Product {
    int id;
    String name;
    double price;
```

```
    static double discountRate = 10;

    void calculateFinalPrice() {
        int i = id;
        String n = name;
        double p = price;

        double discount = (p * discountRate) / 100;
        double finalPrice = p - discount;

        System.out.println("Product ID: " + i);
        System.out.println("Product Name: " + n);
        System.out.println("Price: " + p);
        System.out.println("Discount: " + discountRate + "%");
        System.out.println("Final Price: " + finalPrice);
    }

    public static void main(String[] args) {
        Product p = new Product();

        p.id = 101;
        p.name = "Laptop";
        p.price = 50000;

        p.calculateFinalPrice();
    }
}
```

OUTPUT:-

Product ID: 101

Product Name: Laptop

Price: 50000.0

Discount: 10.0%

Final Price: 45000.0

using local variables.

20) Create a class Bank Account with fields balance, and a class variable bank Code. Use local variables to ensure withdrawal does not exceed balance.

```
class BankAccount {
   double balance;


   static String bankCode = "SBI001";


   void withdraw() {
      double amount = 5000;
      double currentBalance = balance;


      if (amount <= currentBalance) {
         currentBalance = currentBalance - amount;
         balance = currentBalance;


         System.out.println("Bank Code: " + bankCode);
         System.out.println("Withdrawal Amount: " + amount);
         System.out.println("Remaining Balance: " + balance);
      } else {
         System.out.println("Bank Code: " + bankCode);
         System.out.println("Insufficient Balance");
```

```
    }
  }

  public static void main(String[] args) {
    BankAccount b = new BankAccount();

    b.balance = 10000;

    b.withdraw();
  }
}
```

OUTPUT:-

Bank Code: SBI001

Withdrawal Amount: 5000.0

Remaining Balance: 5000.03

21) Create a class Palindrome with a method using local variables to check if a number is palindrome. Add a class variable countChecks.

```
class Palindrome {
  static int countChecks = 0;

  void checkPalindrome(int num) {
    int original = num;
    int reverse = 0;
    int digit;

    while (num > 0) {
      digit = num % 10;
      reverse = reverse * 10 + digit;
```

```
        num = num / 10;
      }

      countChecks++;

      if (original == reverse)
        System.out.println(original + " is a Palindrome");
      else
        System.out.println(original + " is not a Palindrome");
    }

    public static void main(String[] args) {
      Palindrome p = new Palindrome();

      p.checkPalindrome(121);
      p.checkPalindrome(123);

      System.out.println("Total Checks: " + countChecks);
    }
}
```

OUTPUT:-

121 is a Palindrome

123 is not a Palindrome

Total Checks: 2

22) Create a class Armstrong with a method using local variables to check Armstrong number. Add a class variable total Checks.

```
class Armstrong {
    static int totalChecks = 0;
```

```
void checkArmstrong(int num) {
   int original = num;
   int sum = 0;
   int digit;

   while (num > 0) {
      digit = num % 10;
      sum = sum + (digit * digit * digit);
      num = num / 10;
   }

   totalChecks++;

   if (original == sum)
      System.out.println(original + " is an Armstrong number");
   else
      System.out.println(original + " is not an Armstrong number");
}

public static void main(String[] args) {
   Armstrong a = new Armstrong();

   a.checkArmstrong(153);
   a.checkArmstrong(123);

   System.out.println("Total Checks: " + totalChecks);
```

```
    }
}
```

OUTPUT:-

153 is an Armstrong number

123 is not an Armstrong number

Total Checks: 2

23) Create a class Prime Checker with a method using local variables to check prime. Add a class variable total Prime Checks.

```
class PrimeChecker {
    static int totalPrimeChecks = 0;

    void checkPrime(int num) {
        int count = 0;

        for (int i = 1; i <= num; i++) {
            if (num % i == 0)
                count++;
        }

        totalPrimeChecks++;

        if (count == 2)
            System.out.println(num + " is a Prime number");
        else
            System.out.println(num + " is not a Prime number");
    }

    public static void main(String[] args) {
```

```
    PrimeChecker p = new PrimeChecker();


    p.checkPrime(7);
    p.checkPrime(10);


    System.out.println("Total Prime Checks: " + totalPrimeChecks);
  }
}
```

OUTPUT:-

7 is a Prime number

10 is not a Prime number

Total Prime Checks: 2

24) Create a class Fibonacci with a method to display first N terms using local variables. Add a class variable series Count.

```
class Fibonacci {
  static int seriesCount = 0;


  void displaySeries(int n) {
    int a = 0;
    int b = 1;
    int c;


    for (int i = 1; i <= n; i++) {
      System.out.print(a + " ");
      c = a + b;
      a = b;
      b = c;
    }
```

```
    seriesCount++;
    System.out.println();
  }

  public static void main(String[] args) {
    Fibonacci f = new Fibonacci();

    f.displaySeries(7);
    f.displaySeries(5);

    System.out.println("Series Count: " + seriesCount);
  }
}
```

OUTPUT:-

0 1 1 2 3 5 8

0 1 1 2 3

Series Count: 2

25) Create a class Factorial with a method to compute factorial using local variables. Add a class variable total Calls.

```
class Factorial {
  static int totalCalls = 0;

  void calculateFactorial(int n) {
    int fact = 1;

    for (int i = 1; i <= n; i++) {
      fact = fact * i;
```

```
    }

    totalCalls++;

    System.out.println("Factorial of " + n + " = " + fact);
  }

  public static void main(String[] args) {
    Factorial f = new Factorial();

    f.calculateFactorial(5);
    f.calculateFactorial(4);

    System.out.println("Total Calls: " + totalCalls);
  }
}
```

OUTPUT:-

```
Factorial of 5 = 120
Factorial of 4 = 24
Total Calls: 2
```

26) Create a class Matrix with fields for 2x2 matrix and a class variable matrix Type. Use local variables to add and subtract two matrices.

```
class Matrix {
  int[][] a = {{1, 2}, {3, 4}};
  int[][] b = {{5, 6}, {7, 8}};

  static String matrixType = "2x2";
```

```
void calculate() {
    int[][] add = new int[2][2];
    int[][] sub = new int[2][2];

    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            add[i][j] = a[i][j] + b[i][j];
            sub[i][j] = a[i][j] - b[i][j];
        }
    }

    System.out.println("Matrix Type: " + matrixType);

    System.out.println("Addition:");
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            System.out.print(add[i][j] + " ");
        }
        System.out.println();
    }

    System.out.println("Subtraction:");
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            System.out.print(sub[i][j] + " ");
        }
        System.out.println();
```

```
    }
  }

  public static void main(String[] args) {
    Matrix m = new Matrix();
    m.calculate();
  }
}
```

Output:

Matrix Type: 2x2

Addition:

6 8

10 12

27) Create a class StringReverse with a method using local variables to reverse a string. Add a class variable totalReversals.

```
class StringReverse {
  String text = "Hello";

  static int totalReversals = 0;

  void reverse() {
    String reversed = "";

    for (int i = text.length() - 1; i >= 0; i--) {
      reversed = reversed + text.charAt(i);
    }

    totalReversals++;
```

```
    System.out.println("Original String: " + text);
    System.out.println("Reversed String: " + reversed);
    System.out.println("Total Reversals: " + totalReversals);
  }

  public static void main(String[] args) {
    StringReverse s = new StringReverse();
    s.reverse();
  }
}
```

Output:

Original String: Hello

Reversed String: olleH

Total Reversals: 1

28) Create a class Word Counter with a method using local variables to count words in a sentence. Add a class variable language.

```
class WordCounter {
  String language = "English";

  void countWords(String sentence) {
    String[] words = sentence.trim().split("\\s+");
    int count = words.length;

    System.out.println("Sentence: " + sentence);
    System.out.println("Language: " + language);
    System.out.println("Number of words: " + count);
  }
```

```
  public static void main(String[] args) {
    WordCounter obj = new WordCounter();
    obj.countWords("Java is easy to learn");
  }
}
```

OUTPUT:-

Sentence: Java is easy to learn

Language: English

Number of words: 5

29) Create a class Employee with fields name, salary, and a class variable organization. Use local variables to compare salaries of two employees.

```
class Employee {
  String name;
  double salary;
  static String organization = "ABC Company";

  Employee(String n, double s) {
    name = n;
    salary = s;
  }

  void compareSalary(Employee e) {
    double salary1 = salary;
    double salary2 = e.salary;

    System.out.println("Organization: " + organization);
```

```
    if (salary1 > salary2)
      System.out.println(name + " has higher salary.");
    else if (salary2 > salary1)
      System.out.println(e.name + " has higher salary.");
    else
      System.out.println("Both employees have equal salary.");
  }

  public static void main(String[] args) {
    Employee e1 = new Employee("Rahul", 50000);
    Employee e2 = new Employee("Aman", 45000);

    e1.compareSalary(e2);
  }
}
```

OUTPUT:-

Organization: ABC Company

Rahul has higher salary.

30) Create a class Library with fields books Available, and a class variable library Name. Use local variables to update issue and return operations.

```
class Library {
  int booksAvailable;
  static String libraryName = "City Library";

  Library(int books) {
    booksAvailable = books;
  }
```

```
void issueBook() {
    int available = booksAvailable;

    if (available > 0) {
        available--;
        booksAvailable = available;
        System.out.println("Book issued successfully.");
    } else {
        System.out.println("No books available.");
    }
}

void returnBook() {
    int available = booksAvailable;
    available++;
    booksAvailable = available;
    System.out.println("Book returned successfully.");
}

void display() {
    System.out.println("Library: " + libraryName);
    System.out.println("Books Available: " + booksAvailable);
}

public static void main(String[] args) {
    Library obj = new Library(5);
```

```
        obj.display();
        obj.issueBook();
        obj.returnBook();
        obj.display();
    }
}
```

OUTPUT:-

```
Library: City Library
Books Available: 5
Book issued successfully.
Book returned successfully.
Library: City Library
Books Available: 5
```